House Price Prediction

Project Overview

The goal of this project is to build a machine learning model that predicts house prices based on property features such as area, number of rooms, location type, and amenities. The dataset used is the Kaggle Housing Prices Dataset containing 545 records and 13 features. Two regression models were trained and compared — Linear Regression and Random Forest Regressor — using an 80/20 train-test split and 5-fold cross-validation.

Model Performance

Both models were evaluated using MAE, RMSE, R2 Score, and MAPE. Linear Regression slightly outperformed Random Forest on the test set. The Random Forest model overfits on training data (train R2 of 0.95) but generalises similarly in cross-validation, confirming the dataset is too small for tree-based models to gain a significant advantage.

Metric	Linear Regression	Random Forest
MAE	Rs. 9,70,043	Rs. 10,14,947
RMSE	Rs. 13,24,507	Rs. 13,99,769
Test R2	0.6529	0.6124
MAPE	21.04%	21.73%
5-Fold CV R2	0.6324 ± 0.07	0.6203 ± 0.06
Winner	**Linear Regression**	**—**

Key Insights

1. Most Influential Features

Area (plot size in sq ft) is the strongest predictor with a Pearson correlation of +0.54 and a Random Forest importance score of around 0.47, meaning it alone drives nearly half the model's decisions. Bathrooms rank second with a correlation of +0.52 and importance of around 0.15. Airconditioning ranks third. Features like guestroom, hotwaterheating, and mainroad have very low impact and barely affect the predicted price.

2. Model Accuracy in Plain Terms

Both models explain around 63 to 65 percent of price variation. The average prediction error is roughly Rs. 9.7 lakhs on a median home price of Rs. 43.4 lakhs, which is about a 22 percent margin. This is reasonable for a 545-record dataset with no GPS coordinates or neighbourhood quality index. The remaining 35 percent of unexplained variation likely comes from missing features such as exact location, property age, and proximity to amenities.

3. Surprising Findings

Bathrooms turned out to be a stronger price signal than bedrooms, with a correlation of 0.52 versus 0.37. This suggests buyers use bathroom count as a better proxy for home quality than bedroom count. It was also unexpected how much weight airconditioning carries as a simple yes/no feature, indicating buyers in this market place a very high premium on it. Area alone accounts for nearly half the model's predictive power, showing how dominant a single feature can be in this dataset.

4. Business Recommendation

Real estate agents should always highlight area, bathroom count, and airconditioning first in every property listing since these three features drive price more than anything else. For price estimation, comparing homes based on area and bathrooms gives a more accurate benchmark than using bedroom count alone. To improve future model accuracy, the business should start recording property age, floor level, distance to schools and metro stations, and neighbourhood quality scores, as these are the missing factors that currently limit prediction accuracy to around 65 percent.